AgentCompose — Supplementary Material

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This supplementary material supports the main paper *AgentCompose: Quantifying Compositional Generalization in LLM Tool Use* with extended tables, statistical test outputs, error analysis breakdowns, and tool / template inventories. Refer to the main paper for definitions and notation.

1 Per-Agent Per-Split Metric Tables

Table 1 reports TSR, COC, and ABA for all five template agents across all four splits. The headline finding is that ChainAware uniquely achieves TSR = 1.000 on both *val* (in-distribution) and *test_substitutability* (with EquivAware); the other four agents plateau at TSR = 0.25 on substitutability because they never invoke the correct variant named in the natural-language prompt.

2 DeepSeek-V4-Flash Validation

Bootstrap 95% confidence intervals (1000 iterations on per-task TSR) have half-widths ≤ 0.10 on every cell at N=100, confirming the qualitative pattern is statistically stable: even frontier 2026 LLMs hit near-zero TSR on productivity.

3 Statistical Test Outputs

3.1 Friedman Test

3.2 Critical Difference

Refer to Figure **??** in the main paper. The CD threshold at $\alpha = 0.05$ for $k = 5$ algorithms and N tasks is 1.477 (Demsar 2006, Table 5). Agents whose mean ranks differ by more than 1.477 are significantly different.

Agent	Split	N	TSR	COC	ABA
ReAct	val	400	0.268	0.125	0.057
	test_systematicity	600	0.428	0.060	0.055
	test_productivity	600	0.542	0.088	0.062
	test_substitutability	200	0.250	0.000	0.057
Native FC	val	400	0.268	0.125	0.284
	test_systematicity	600	0.145	0.060	0.195
	test_productivity	600	0.200	0.000	0.196
	test_substitutability	200	0.250	0.000	0.297
Reflexion	val	400	0.268	0.125	0.057
	test_systematicity	600	0.428	0.060	0.055
	test_productivity	600	0.542	0.088	0.062
	test_substitutability	200	0.250	0.000	0.057
Plan-Execute	val	400	0.268	0.125	0.282
	test_systematicity	600	0.145	0.060	0.208
	test_productivity	600	0.107	0.000	0.213
	test_substitutability	200	0.250	0.000	0.292
ChainAware (Ours)	val	400	**1.000**	**1.000**	0.220
	test_systematicity	600	0.438	0.335	0.155
	test_productivity	600	0.452	0.000	0.178
	test_substitutability	200	**1.000**	0.205	0.221

Table 1: Per-agent per-split metric table (TSR, COC, ABA). ChainAware+EquivAware fully closes val and substitutability; other splits remain challenging across all agents.

3.3 Pairwise Wilcoxon (selected contrasts)

4 Error Analysis Breakdown

5 EquivAware Ablation

EquivAware is a deterministic, parameter-free overlay on ChainAware. We quantify its isolated effect on substitutability:

6 Tool Library (30 tools across 8 domains)

- **Information Retrieval** (5): search_web, search_news, fetch_url, summarize, translate

Agent	Val	Sys	Prod	Sub
DeepSeek-ReAct (N=100)	0.130	0.060	0.040	0.140
DeepSeek-FC (N=100)	0.240	0.080	0.070	0.210

Table 2: DeepSeek-V4-Flash per-split TSR at N=100. All cells complete; 95% bootstrap CIs in Table 3.

Agent / Split	TSR	95% CI	CGG	95% CI
DeepSeek-ReAct val	0.130	[0.060, 0.210]	—	—
DeepSeek-ReAct sys	0.060	[0.020, 0.110]	0.070	[−0.020, 0.170]
DeepSeek-ReAct prod	0.040	[0.010, 0.080]	0.090	[0.000, 0.190]
DeepSeek-ReAct sub	0.140	[0.070, 0.220]	−0.010	[−0.130, 0.110]
DeepSeek-FC val	0.240	[0.160, 0.320]	—	—
DeepSeek-FC sys	0.080	[0.030, 0.140]	0.160	[0.050, 0.260]
DeepSeek-FC prod	0.070	[0.030, 0.120]	0.170	[0.070, 0.270]
DeepSeek-FC sub	0.210	[0.130, 0.300]	0.030	[−0.090, 0.150]

Table 3: DeepSeek-V4-Flash TSR + CGG with 95% bootstrap confidence intervals (1000 iterations on per-task TSR). CIs are reported in CGG only when statistically distinct from zero.

- **Computation** (5): calculator, unit_convert, statistics, round, sort
- **Communication** (4): send_email, send_sms, send_push, send_slack
- **File I/O** (5): read_file, write_file, append_file, list_files, delete_file
- **Scheduling** (4): get_datetime, calendar_add, calendar_list, format_date
- **Data / Database** (4): sql_query, csv_filter, db_insert, json_transform
- **Image / Vision** (4): image_resize, image_crop, image_classify, ocr_extract
- **Audio / Speech** (3): audio_transcribe, text_to_speech, audio_classify

Six equivalence groups span these tools:

- G1_search: `search_web` ≡ `search_news`
- G2_communicate: `send_email` ≡ `send_sms`
- G3_data_query: `sql_query` ≡ `csv_filter`

Split	χ^2	df	p-value
test_systematicity	41.7	4	< 0.001
test_productivity	87.4	4	< 0.001
test_substitutability	246.0	4	< 0.001

Table 4: Friedman test across 5 template agents, per-task paired ranks. Null hypothesis (all agents rank equally) rejected on every split.

Split	Contrast	W	p_{uncorr}	Cohen's d
substitutability	ChainAware+EquivAware vs ChainAware	—	< 0.001	-2.44
substitutability	ChainAware+EquivAware vs Native FC	—	< 0.001	-2.44
productivity	ChainAware vs Native FC	—	< 0.001	-0.65
systematicity	ChainAware vs Native FC	—	< 0.001	-0.78

Table 5: Selected pairwise Wilcoxon signed-rank contrasts. Full 5×5 matrix in `results/statistical_tests.json` (attached as supplementary archive).

- G4_visual_recognize: `image_classify` ≡ `ocr_extract`
- G5_notify: `send_slack` ≡ `send_push`
- G6_calendar_op: `calendar_add` ≡ `calendar_list`

Full JSON schemas for each tool are in `tools/*.json` (released with the code archive).

7 Task Templates (25 across 6 chain lengths)

The 25 task templates span chain lengths 2–5 and cross all 8 domains. Full YAML in `templates/task_templates.yaml`. Examples:

- *T1*: `search_web` → `send_email` — "Look up the weather in Paris and email it to alice@example.com."
- *T6*: `calculator` → `send_sms` — "Compute 42×17 and SMS the result to +1-555-0100."
- *T11*: `search_web` → `statistics` → `write_file` — "Fetch the last 30 days of AAPL closing prices, compute mean and stdev, and save as CSV."

Error type	Val	Sys	Prod	Sub
missing_tool	0.73	0.86	0.80	1.00
wrong_tool	0.10	0.05	0.10	0.00
wrong_order	0.05	0.05	0.05	0.00
bad_argument	0.10	0.04	0.05	0.00

Table 6: Error-type breakdown for Native FC (rates among failed tasks). Missing-tool is the dominant failure mode across every split; 100% of substitutability failures are missing-tool.

Variant	Substitutability TSR	Cohen's d vs baseline
ChainAware (baseline)	0.250	—
ChainAware + EquivAware (Ours)	**1.000**	-2.44

Table 7: EquivAware lifts ChainAware's substitutability TSR from 0.250 (constant-tool baseline) to 1.000; no other split is affected.

- *T18*: `audio_transcribe` → `summarize` → `send_email` — "Transcribe `meeting.mp3`, summarize the action items, and email them to the team."

8 Hyperparameters and Configuration

- Random seed: 42 (default in all scripts)
- `max_steps`: 10 per task
- `n_synthetic_chains` (ChainAware): 50
- Bootstrap iterations: 1000
- Friedman / Wilcoxon α: 0.05 with Bonferroni correction
- DeepSeek temperature: 0 (deterministic)
- DeepSeek `max_tokens`: 1024 per turn

9 Bootstrap 95% Confidence Intervals (TSR)

1000 resamples, seed=42. Full per-agent values in `results/bootstrap_ci.json`

Variance across independent training runs is not reported because all template agents are deterministic under the fixed seed 42; the bootstrap

Agent	Val	Sys.	Prod.	
ReAct	0.268 [0.225, 0.312]	0.428 [0.390, 0.467]	0.542 [0.503, 0.580]	0.
Native FC	0.268 [0.225, 0.312]	0.145 [0.117, 0.173]	0.200 [0.168, 0.233]	0.
Reflexion	0.268 [0.225, 0.312]	0.428 [0.390, 0.467]	0.542 [0.503, 0.580]	0.
Plan-Execute	0.268 [0.225, 0.312]	0.145 [0.117, 0.173]	0.107 [0.083, 0.133]	0.
ChainAware	1.000 [1.000, 1.000]	0.438 [0.400, 0.477]	0.452 [0.412, 0.492]	1.
DeepSeek ReAct	0.130 [0.060, 0.200]	0.060 [0.020, 0.110]	0.040 [0.010, 0.080]	0.
DeepSeek FC	0.240 [0.160, 0.320]	0.080 [0.030, 0.140]	0.070 [0.030, 0.120]	0.

Table 8: Task Success Rate (TSR) with bootstrap 95% CIs per split.

CIs above capture task-sampling variance. Chain-length-stratified success rates are deferred to the released raw per-task records (regenerable via `scripts/run_all.py -skip-llm`).

10 Reproducibility Checklist (NeurIPS-style)

- ✓ Code released (MIT license)
- ✓ Data released (3,000 tasks, 25 templates)
- ✓ Simulator released (deterministic JSON API)
- ✓ Random seeds documented
- ✓ Hyperparameters documented
- ✓ One-command reproduction (`python scripts/run_all.py -skip-llm`)
- ✓ Test suite (39/39 passing in $<$30 s)
- ✓ LLM API costs disclosed
- ✓ Statistical tests with effect sizes